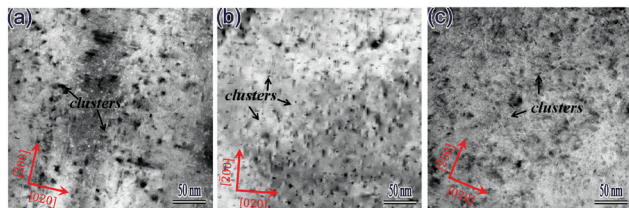


Advanced aluminium alloy for thermal stability in EV

Researchers from the Department of Aluminum in the Advanced Metals Division at KIMS have developed a new aluminium alloy for electric vehicles that enhances thermal stability by up to 140%, as published in the Journal of Materials Research and Technology. The alloy uses nanostructures



HRTEM images of the (a) Base alloy, (b) Ag-added alloy, and (c) Ge-added alloy aged at 200°C for 2 h (Credit: Journal of Materials Research and Technology)

and trace elements in the 6000 series aluminium to prevent thermal degradation, improving safety as vehicles age. This innovation addresses the growing electric vehicle battery enclosure material market, projected to reach US\$863.3 million by 2029. The new alloy's superior thermal properties also make it suitable for supersonic aircraft, with ongoing research focused on industrial applications and export potential.

Advances in greener, cheaper EV battery technology

The global shift to electric vehicles (EVs) is gaining momentum, but battery material extraction has significant environmental and cost concerns. Research from McGill University presents advancements in creating more affordable and eco-friendly



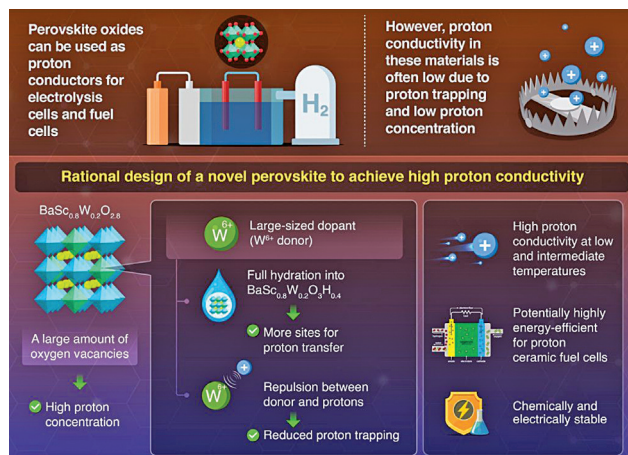
Richie Fong, a PhD student in Materials Engineering, conducts research on cathodes in a McGill lab (Credit: McGill University)

lithium-ion batteries for EVs. The studies focus on disordered rock-salt-type (DRX) cathode materials, using sustainable and cost-effective metals. PhD student Richie Fong's study, published in Advanced Energy Materials, explores iron-based DRX cathodes, achieving unprecedented storage capacities and potentially reducing battery costs by 20%. Another study,

in Energy & Environmental Science by Prof. Jinhyuk Lee, investigates manganese-based DRX cathodes. By adding multi-walled carbon nanotubes and an adhesive binder, the team achieved the highest energy density for Mn-DRX cathodes, promising more affordable and sustainable energy storage. An industry partner is already collaborating to market these innovations.

High proton conductivity perovskite for fuel cells

Researchers at Tokyo Institute of Technology (Tokyo Tech), led by Professor Masato Yashima, have developed a perovskite material, BaSc_{0.8}W_{0.2}O_{2.8}, which exhibits excellent proton conductivity at low and intermediate temperatures. This advancement promises to enhance the efficiency of protonic ceramic fuel cells (PCFCs) and proton-conducting



High proton conduction by full hydration in highly oxygen deficient perovskite (Credit: <https://www.kek.jp/en/press>)

electrolysis cells (PCECs). The perovskite's high concentration of intrinsic oxygen vacancies and incorporation of large W⁶⁺ ions as dopants allow for greater water absorption, increasing proton concentration and reducing proton trapping. This development addresses the current limitations of proton conductors, providing a pathway for designing high-performance materials for PCFCs and PCECs. This research lays the groundwork for more efficient energy conversion technologies.

Digital twin boosts manufacturing speed and quality

Researchers at the University of Michigan have developed a method to optimise manufacturing machine speeds while maintaining quality. Using digital twin technology, the new algorithm automates feed rate adjustments, eliminating the need for trial and error or conservative approaches that